W03

Statistics foundations for ecologists

STATISTICS

Most tests taught separately (t-test, ANOVA, regression) are the same model: the linear model. Today you describe data, quantify uncertainty with confidence intervals and fit linear models to bird abundance.

DURATION	About 2 hours.
YOU NEED	R with the tidyverse and your course project.
FILES	`week03_statistics_foundations.R`, `week03_solutions.R`, `survey_points.csv`, `bird_counts_wide.csv`
DATA	Richness and abundance per point, calculated as in Week 2.

Learning outcomes

- tell the standard deviation from the standard error;
- calculate confidence intervals with the t distribution and by bootstrapping;
- compare groups with `t.test()` and `lm()`, and see they agree;
- fit and interpret simple and multiple linear regression;
- check model assumptions with residuals.

1 · Description and uncertainty

The **standard deviation** (SD) describes how much points vary. The **standard error** (SE = SD / square root of n) describes uncertainty in the *mean*: it shrinks as sample size grows. A 95% confidence interval is roughly the mean ± 2 SE.

Abundance (birds per visit)	Value
Mean	9.85
Median	10.0
Standard deviation	3.72
Standard error	0.59
95% CI (t)	8.66 – 11.04
95% CI (bootstrap)	8.71 – 11.00

```
set.seed(1)
boot_means <- replicate(5000, mean(sample(x, replace = TRUE)))  # resample points
quantile(boot_means, c(0.025, 0.975))                            # 95% CI
```

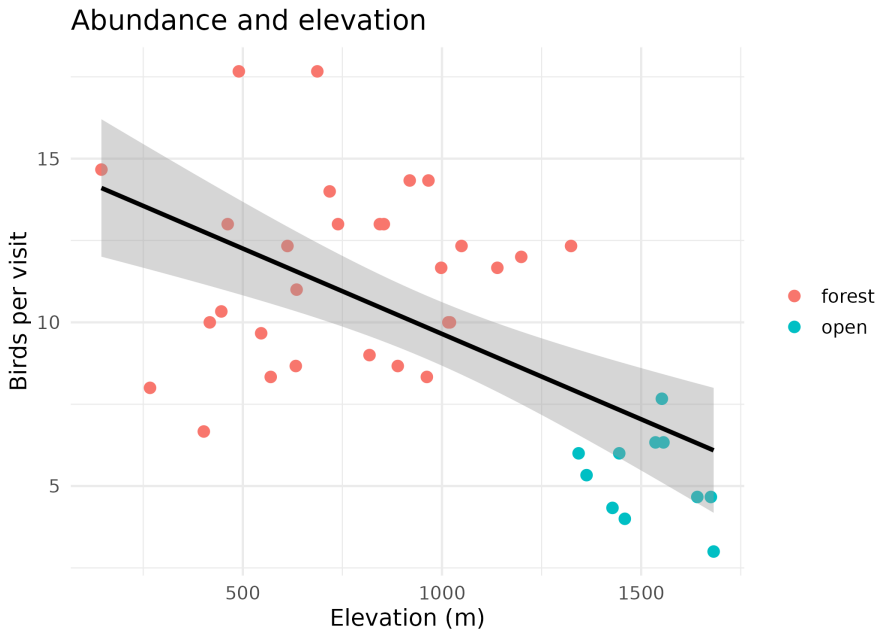
2 · Comparing groups is a linear model

Forest versus open habitat: the Welch t-test gives a difference of 6.3 birds per visit (95% CI 5.0 to 7.6; p < 0.001). `lm(abundance ~ open)` gives exactly that difference as a coefficient. With more than two groups, `anova(lm(abundance ~ habitat))` is the classic ANOVA (F = 12.4, p < 0.001), and `TukeyHSD()` runs pairwise comparisons corrected for multiple testing.

> **How to read factor coefficients**
>
> R uses the first level (alphabetically) as the reference. The intercept is that group's mean, and each coefficient is another group's difference from it. Change the reference with `relevel()` or `fct_relevel()`.

3 · Linear regression

`lm(abundance ~ elevation_m)`: abundance falls by 0.52 birds per visit for every 100 m climbed (p < 0.001), and elevation explains 36% of the variation (R^2 = 0.36). `predict(..., interval = "confidence")` gives the expected abundance at a given elevation with its interval.



Abundance against elevation with the regression line and its 95% confidence interval.

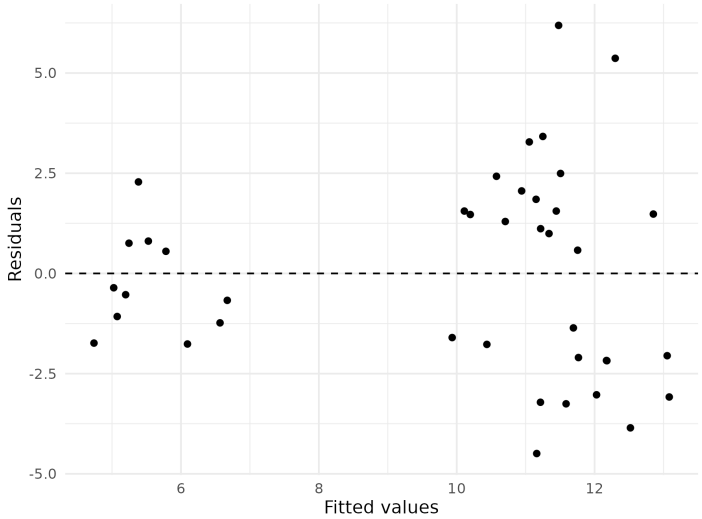
Multiple regression and correlated predictors

Elevation and canopy cover are correlated (r = -0.69). With both in the model, canopy cover explains abundance (p < 0.001) and the elevation effect is no longer significant: much of the "elevation effect" was really forest loss. AIC drops from 206 to 193, so the two-predictor model is better. Strongly correlated predictors make coefficients unstable; Week 10 looks at choosing variables.

4 · Checking assumptions

`plot(model)` draws four diagnostic plots. The essentials:

- **Residuals vs fitted:** a shapeless cloud, no curve, no funnel.
- **Q-Q:** points close to the line (roughly normal residuals).
- **Residuals vs leverage:** no point with Cook's distance above 1.



Residuals against fitted values for the elevation + canopy model.

> **A warning for Week 4**
>
> Richness and counts are whole numbers of zero or more, and their variance usually grows with the mean. A linear model can predict negative values and assumes constant variance. For these data we use generalised linear models (GLMs).

Exercises

Answers in `week03_solutions.R`.

1. Mean, median and 95% CI (t and bootstrap) of richness. Do they agree?
2. Does canopy cover differ between forest and open points? Check that `t.test()` and `lm()` give the same difference.
3. Fit richness against elevation. How much does it change per 100 m? What is the R^2?
4. Predict richness at 1,800 m. Does it make sense? Why is extrapolation risky?
5. Check the residuals of the habitat model. Do you see a problem?

Before Week 4

- Install `install.packages(c("MASS", "DHARMa", "lme4"))`.
- Optional reading: chapters 2 and 3 of *Beckerman, Childs and Petchey, Getting Started with R*, or any introduction to the linear model.